

Determining Safest Routes in San Francisco Using The Zero-Inflated Negative Binomial Regression Model

Huy H. Le¹, J.D. Bunker², Rok Jun Lee², Narae Wadsworth²

¹Author; ² Collaborator – Master’s Candidate.

Department of Statistics, University of California, Berkeley.

{lehuy, Jd.bunker, rokjunlee, nwadsworth}@berkeley.edu

Abstract. *In search of determining the safest routes to travel in San Francisco, the naive/trivial criterion is to protect yourself from crimes by avoiding them. Thus, this paper proposes using a zero-inflated negative binomial model to assign risk scores for regions of San Francisco. By splitting the map of San Francisco into a grid consists with 0.1 miles unit squares, and aggregating numbers of weighted-crimes over 16 years (2003 – 2018), we were able to come up with a reasonable framework that suggests routes with lowest chance of encountering crimes.*

Keywords: *count data, negative binomial, zero-inflated.*

1. Introduction & Motivation

San Francisco is one of the most developed cities in the world and home of many tech companies. As a result, the population of San Francisco has been increasing for the past years, and although this created a diverse community, the crime rate has also been growing significantly. Hence, this paper discusses how to determine the safest route to go from any point A to B in San Francisco based on avoiding crimes criterion. Safety here is determined by how often crimes happen on a given path and how severe the crime is. The data is provided by the San Francisco Police Department and recorded from 2003 to 2018, which can be found on DataSF website. The goal of this paper is to establish an algorithm that is useful for people who live in or travel to San Francisco as it can effectively and efficiently lower the risk of running into crimes.

2. Data & Preliminary Results

2.1. Data Description

The data set is originally from the San Francisco Police Department and then provided by a group name DataSF. The data has over two million rows and 13 covariates, which present the information about each crime that happened in San Francisco including date and time that the crime occurred (including month, day, hour, minute), type of crimes (39 different types), locations (address, latitude, longitude, and district), incident number. Specifically, the 13 variables of interest are: *incident number, category, description, day of week, date, time, pd district, resolution, address, latitude, longitude, location, pd ID.*

Before we do any analysis, we filter out the crime’s categories that are not relevant such as trespassing, fraud, etc. As a result, we end up with only 24 different types of crimes. We then introduce the weight for each type of crimes since the severity of each crime should be treated differently. Therefore, using the seriousness scores developed by the Bureau of Justice Statistics and Florida Department of Law Enforcement, we are able to assign the weight to each type of crimes for the whole data set.

2.2. Exploratory Data Analysis (EDA)

We, first, factorized some covariates such as incident number, category, day of week, hour, district. Some problem we found with the data set is that there are multiple repeated incident numbers since a crime belongs into different crime types. Moreover, there are few crimes have coordinates that do not belong to San Francisco. As a result, we removed these wrong-coordinated crimes, and for the repeated ID case, we add up the weights of this crime and then remove all the duplicated so that there is only one observation with this ID remains. After this process, we are now ready to perform EDA. For the preliminary step, we plot the density of top six crimes in San Francisco.

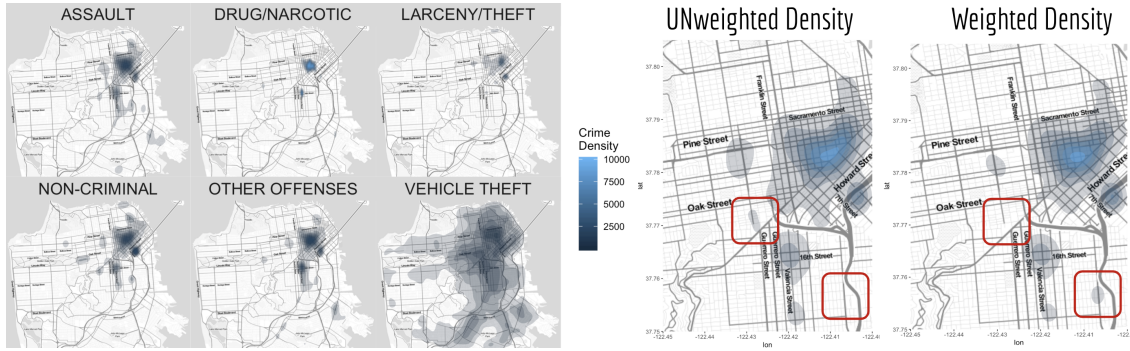


Figure 1. Density plots of top six crimes (left) and Unweighted versus Weighted Density (right).

Looking at **Figure 1**, we can see that these crimes mostly concentrated in between Sacramento Street and Howard Street, although vehicle theft seems to have a broader range. Additionally, Drug/Narcotic has higher concentration at one location while the other top crimes are more spread out. After that, using the severity scores discussed above, we plot the weighted density estimation version of the crime data (**Figure 2**). Consequently, the distribution slightly shifts and has altered shape; however, the distribution is still somewhat similar. As a result, we see that adding weights into the data does not change the distribution significantly and yet gives more accuracy about safety. Thus, we are confident in using these seriousness scores to do further analysis to determine safety.

3. Logistics & Methodology

3.1. Constructing Unit-Squared Grid

Several methods can be used to break down San Francisco's map to determine the crime rate of San Francisco and how it is distributed. One option is to use street intersections data of San Francisco and implement a graphing algorithm, in particular, edge-weighted graphs. However, we decided that this approach is not efficient as we do not want to suggest a 2-hour route over a 5-minute walk simply because it has a lower risk score than the 5-minute one. Additionally, we need to know the crime rate for each street in San Francisco to use this graphing algorithm. However, the crime data is not sufficient to determine this quantity since crimes are reported by either hundred of blocks or intersections where crimes happened. Thus, we decide to break down the map into 0.1-mile square boxes and obtain weighted crime frequencies for each box.

The choice of 0.1 miles (160 m) is determined by balancing the accuracy of the desired crime frequency in each unit, and the average length of a city block in the United States is about

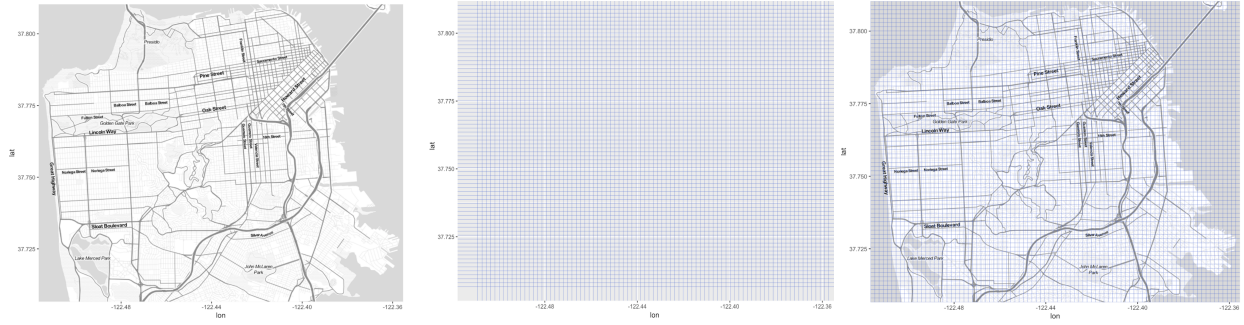


Figure 2. Applying the unit-squared grid onto the map of San Francisco.

660 feet (200 m). Then, this 0.1-mile grid is converted into latitude and longitude to be consistent with the data. Next, we assign the severity score to each box by using the weights of crimes and averaging over the months and years. Specifically, let s_i be the severity score of each unit square, then

$$s_i = \begin{cases} 1/(|Y|) & \text{if } \sum_{Y=1}^{|Y|} y_{(i|Y)} = 0 \\ 1/(|Y|) \sum_{Y=1}^{|Y|} y_{(i|Y)} w_{(i|Y)} & \text{otherwise} \end{cases},$$

where $|Y| = 16$: number of years, $y_{(i|Y)}$: is the crime counts of the i^{th} box in a given year, and $w_{(i|Y)}$: is the seriousness weight corresponding to $y_{(i|Y)}$.

3.2. Zero-Inflated Negative Binomial Regression (ZINB)

When considering the nature of the data, it is easy to notice that crime data is count data, and thus, either Poisson and Negative Binomial Regression are good candidates for this type of data. Also, after performing exploratory data analysis, we discovered that there is the presence of considerable variability in a data set, i.e., overdispersion, which the Poisson regression model does not account for. As a result, the Negative Binomial model turns out to be a better model in this case. Also, due to the nature of crimes, many areas have no crime reported in these 16 years. In fact, from the data section, the Mission District is the area with most of crimes reported, and there are much fewer crimes reported in other areas of San Francisco. Hence, there is a large number of boxes that have zero crime frequencies, and therefore, suggest the use of Zero-Inflated Negative Binomial is necessary and preferable.

We will discuss on how to employ the ZINB in this study using the results presented in the books by Cameron and Trivedi (2013) and Hilbe (2014) and in Garay, Hashimoto, Ortega, and Lachos (2011). First, the ZINB model consists with two different processes: the first one follows a binary distribution that generates structural zeros, and the second one is the Negative Binomial Distribution that generates counts and possibly zeros as well. Then, we define the following quantities:

- α is the ancillary or heterogeneity parameter.
- $\mu_i := \exp\{x_{i,[1]}^\top \beta\}$, where β 's are the coefficients and $x_{i,[1]}$ is the set of covariates used for the count model. (Latitude and Longitude)
- $\pi_i := \text{expit}\{x_{i,[2]}^\top \gamma\} \Leftrightarrow \text{logit}\pi_i = x_{i,[2]}^\top \gamma$, where γ 's are the coefficients and $x_{i,[2]}$ is the set of covariates used for the zero-inflated model. (District)
- $\{x_i\}_{i=1}^n \in \mathbb{R}^3$ represents three covariates: Latitude, Longitude, and District.
- $\{y_i|x_i, \alpha\}_{i=1}^n$ is the crime frequencies for each unit square given the covariates x_i and α .

Let t be the time the user wants to travel, then probability distribution of the ZINB model is:

$$\mathbb{P}\{y_i^{(t)} = k\} = \begin{cases} \pi_i + (1 - \pi_i)g(y_i^{(t)} = 0) & \text{if } k = 0 \\ (1 - \pi_i)g(y_i^{(t)} = k) & \text{otherwise} \end{cases},$$

where $g(y_i^{(t)}) = \mathbb{P}(Y = y_i^{(t)} | \mu_i, \alpha) = \frac{\Gamma(y_i^{(t)} + \alpha^{-1})}{\Gamma(\alpha^{-1})\Gamma(y_i^{(t)} + 1)} \left(\frac{1}{1 + \alpha\mu_i}\right)^{\alpha^{-1}} \left(\frac{\alpha\mu_i}{1 + \alpha\mu_i}\right)^{y_i^{(t)}}$. Then, the regression coefficients are estimated using the method of maximum likelihood. The log-likelihood function is given as follow: $\ell(\mu_i, \alpha | y_i^{(t)}) = \ell_1 + \ell_2 + \ell_3 - \ell_4$, where

- $\ell_1 = \sum_{\{i: y_i^{(t)}=0\}} \ln \left[\exp\{x_{i,[2]}^\top \gamma\} + (1 + \alpha\mu_i)^{-\alpha^{-1}} \right]$
- $\ell_2 = \sum_{\{i: y_i^{(t)}>0\}} \left[\sum_{j=0}^{y_i^{(t)}-1} \ln(j + \alpha^{-1}) \right]$
- $\ell_3 = \sum_{\{i: y_i^{(t)}>0\}} \left[-\ln(y_i^{(t)}!) - (y_i^{(t)} + \alpha^{-1}) \ln(1 + \alpha\mu_i) + y_i^{(t)} \ln(\alpha) + y_i^{(t)} \ln(\mu_i) \right]$
- $\ell_4 = \sum_{i=1}^n \ln \left(1 + \exp\{x_{i,[2]}^\top \gamma\} \right)$

Let $r \in \{1, 2\}$, $m = 1$, the score function $\nabla \ell$, then is:

$$\begin{aligned} \frac{\partial}{\partial \beta_r} \ell &= \sum_{\{i: y_i^{(t)}=0\}} \left[\frac{-\mu_i(1 + \alpha\mu_i)^{-1-\alpha^{-1}}}{\exp\{x_{i,[2]}^\top \gamma\} (1 + \alpha\mu_i)^{-\alpha^{-1}}} \right] x_{i,[1],r} + \sum_{\{i: y_i^{(t)}>0\}} \left[\frac{y_i^{(t)} - \mu_i}{1 + \alpha\mu_i} \right] x_{i,[1],r} \\ \frac{\partial}{\partial \gamma_m} \ell &= \sum_{\{i: y_i^{(t)}=0\}} \left[\frac{\exp\{x_{i,[2]}^\top \gamma\}}{\exp\{x_{i,[2]}^\top \gamma\} + (1 + \alpha\mu_i)^{-\alpha^{-1}}} x_{i,[2],m} \right] - \sum_{i=1}^n \frac{\exp\{x_{i,[2]}^\top \gamma\}}{1 + \exp\{x_{i,[2]}^\top \gamma\}} x_{i,[2],m} \\ \frac{\partial}{\partial \alpha} \ell &= \sum_{\{i: y_i^{(t)}=0\}} \frac{\ln(1 + \alpha\mu_i)^{(1+\alpha\mu_i)} - \alpha\mu_i}{\alpha^2(1 + \alpha\mu_i) \left[\exp\{x_{i,[2]}^\top \gamma\} (1 + \alpha\mu_i)^{\alpha^{-1}} + 1 \right]} \\ &\quad + \sum_{\{i: y_i^{(t)}>0\}} \left\{ \sum_{j=0}^{y_i^{(t)}-1} \frac{-1}{\alpha^2 j + \alpha} + \frac{\ln(1 + \alpha\mu_i)}{\alpha^2} + \frac{y_i^{(t)} - \mu_i}{\alpha(1 + \alpha\mu_i)} \right\} \end{aligned}$$

The gradient is rather long and complicated, hence it is omitted from this paper. However, it can be found in the books by Cameron and Trivedi (2013) and Hilbe (2014) and in Garay, Hashimoto, Ortega, and Lachos (2011). After that, we use this model to predict the crime counts for each unit square $\hat{y}_i$. Then, we calculate the risk score for each box using the following formula $B_i = s_i \hat{y}_i$, where s_i is the severity score of each unit square.

3.3. Path-Finding Algorithm.

As we finish developing the method to determine the risk score for each unit square, we then need to come up with a way to calculate the risk score for the whole traveling path. In this study, we use the package in R called “googleway” to determine the routes. This package is the R version of Google Maps, which the user inputs the starting and ending points, departure time

(using POSIXct time format – “YYYY-MM-DD hour:minute:second time-zone”) and the results in the JSON file consists with all the information and travel options. The advantage of this package is that it provides the coordinates of the location when the user need to make a turn, and the question of how to calculate the risk score of the whole path is easy to solve.

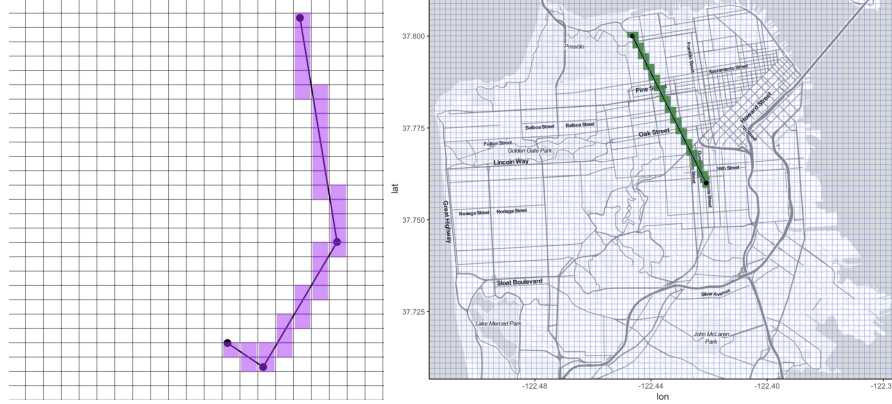


Figure 3. Cartesian coordinates system (left) and San Francisco map (right).

Roughly speaking, the algorithm sequentially uses coordinates of starting and ending points of each path, connects them and determines which boxes the segment crosses. Then, we both add up the boxes and penalize the time spending for each segment (dividing the time spending on each line) to get the final risk score. Thus, we can rephrase the question into finding boxes, and the algorithm is given below.

Box-finding algorithm:

```

1  input (x_0,y_0) and (x_1,y_1);
2  calculate slope (m), intercept (b) from y = mx + b;
3  if (abs(m) < 1) {
4      1. Traverse in the x direction;
5      2. I = intersections of the line and each vertical grid lines;
6      3. h_1, h_2 = min(dist(I,h_i)) where h_i = horizontal lines;
7      4. Identify the box B;
8  } else {
9      1. x <- y, y <- x;
10     2. Follow the same process above;
11 }
12 return B;
```

Table 1. Box-finding algorithm.

Finally, with N is the number of segments from point A to B, T_i is the time spending on each segment, K is the number of boxes that the i^{th} segment crosses through, the total risk of the route is $R = \sum_{i=1}^N (1/T_i) \sum_{j=1}^K B_{i,j}$.

4. Implementation & An Example

After discussing the logistics and methodology of our method, we establish a framework to determine the safest route to travel in San Francisco by demonstrating with an example. After performing EDA, we discovered the time frame has most crime frequencies is Friday, 6 pm, downtown area. Consequently, we subset this the data to this time frame to implement our algorithm.

4.1. Ad Hoc Testing

After subsetting the data, we end up with 1064 unit squares and 114 boxes of zero counts. Then, we begin by comparing four different models Poisson, Negative Binomial, and Zero-Inflated Negative Binomial. First, we performed the dispersion test on the Poisson Regression model by using “dispersiontest” function from “AER” package in R, and there is, in fact, overdispersion in the data. Next, to decide between Negative Binomial and Zero-Inflated Negative Binomial Regression Models, we used the Vuong’s closeness test to examine how different the two models, in R, function “vuongtest” in package “nonnest2”. A quick remark is that both models behave similarly in term of that both give similar results.

Vuong’s Closeness Test (Nested Model)	
Model 1:	zeroinf(formula = Count ~ Longitude + Latitude District)
Model 2:	glm.nb(formula = Count ~ Longitude + Latitude + District)
Variance test:	
	H1: Model 1 and Model 2 are distinguishable
	w2 = 0.017, p = 1.73e-05
Robust likelihood ratio test of distinguishable models:	
	H1: Model 1 fits better than Model 2
	LR = 22.274, p = 1.55e-08

Table 2. Vuong’s Closeness Test Model Results.

As a result, there is a significant difference between the two models, and since the Negative Binomial Model is nested in the Zero-Inflated Negative Binomial Model, hence we choose the ZINB to predict the crime counts.

4.2. Results

After running the Zero-Inflated Negative Model, the results are reasonable in the sense that Latitude and Longitude are significant variables and only some districts matter for our model (**Table 3**). Then, using the “predict” function in R, we obtain the predict values of 1064 boxes of the model. Multiplying each box with the severity score, we are able to get the risk score for each box.

	Estimate	Std. Error	z-value	p-value
Latitude	0.091	0.03363	2.695	0.00704
Longitude	0.49	0.03316	14.667	< 2e-16
Central (D)	-1.9376	0.5354	-3.619	0.000296
Mission (D)	-15.5344	252.1787	-0.062	0.950881
Northern (D)	-15.6185	247.4960	-0.063	0.949682
Park (D)	-15.0092	248.8718	-0.060	0.951910
Richmond (D)	-16.1868	742.0033	-0.022	0.982596
Southern (D)	-2.3037	0.6510	-3.539	0.000402
Tenderloin (D)	-12.1579	200.5600	-0.061	0.951662

Table 3. Results of ZINB model.

The next step is to look at the residual plot, **Figure 4**, to see how good this model performs such as how good does the ZINB model fits the data and any issue with outliers. We can see that most of the residuals are close to the 0 line, which informs the model fits reasonably. There exists an outlier that has extreme values, which has the latitude from 37.77483 to 37.77629, and longitude from -122.4037 to -122.4019. By checking on the map, this is where the San Francisco County Jail located, and thus, it makes sense that a significantly high number of crimes are reported at this location.

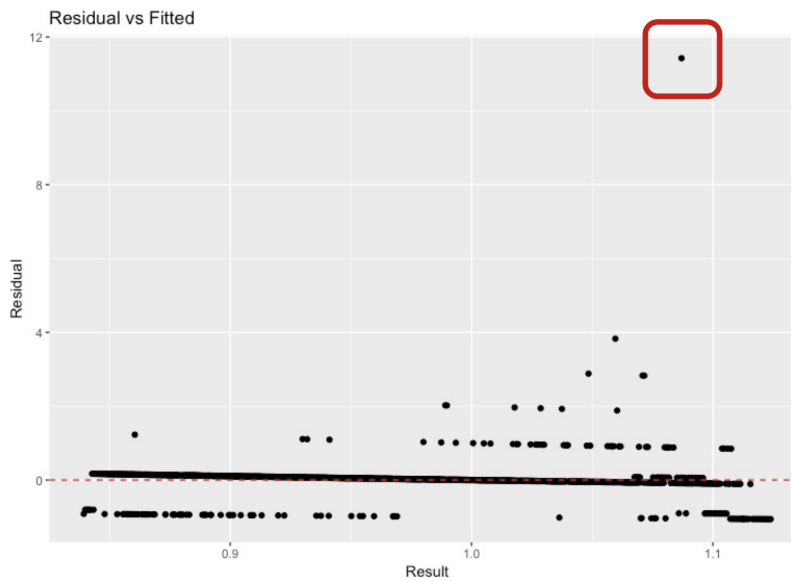


Figure 4. Residual versus Fitted plot.

4.3. Implementation

After implementing the model and able to get the risk score for 1064 boxes, we now can determine the total risk to go from any two places in the San Francisco's downtown area. In this paper, we choose the starting point is "24 Hour Fitness, San Francisco" and ending point is "Civic Center, San Francisco" on Friday 6 pm, Friday, May 10th, 2019, as an example. **Figure 5** presents the results given by our algorithm. Google Maps provide three different options to go between these two locations, and we were able to assign the total risk scores to these three paths. Notice that the higher the risk score, the more red the box is. Therefore, according to our algorithm, the leftmost path is the most ideal route to go from "24 Hour Fitness" to "Civic Center" with the risk score of 0.622.



Figure 5. Options going from "24 Hour Fitness" to "Civic Center", Friday, 6 pm, 05/10/2019.

5. Conclusion & Future Applications

This paper has established a general framework to calculate the risk of encountering crimes in San Francisco. We demonstrated that by breaking down San Francisco's map into 0.1-mile squares, applying the ZINB Regression Model, and box-finding algorithm, we were able to calculate the risk of the traveling path. Moreover, the algorithm took around 3-5 seconds to run, which includes calculating the risk score for all 1064 boxes of the downtown area. This study is significant we were able to demonstrate to at least estimating the chance of encountering crimes reasonably.

Despite the result of this algorithm, there still exist areas for continued development. The most important and crucial aspect is the model we are using. The ZINB we implemented is still a linear model, which District only makes a shift between models. The more desired model is one that can account for the non-linearity of the spatial aspect of the data, and one of the methods that can be integrated into this model to achieve this goal is using K-Nearest Neighbor approach. Moreover, for future research, we want to include additional data sets such as streetlight locations and demographic info to derive more accurate result in determining safety routes in San Francisco. Another improvement that can be made is to make this algorithm more user-friendly since everything is developed in R, and we want this algorithm to be useful.

To sum up, even though the algorithm developed in this study behaves nicely, it still needs to be optimized and improved to achieve the produce more accurate and faster results.

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